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Laboratory of General Anatomy. Academic year 2025-2026.
Course intended for first-year medical students.

GENERAL MYOLOGY

PLAN
I. **INTRODUCTION**
II. CHARACTERISTICS
III. CLASSIFICATION
IV. EXTERNAL CONFIGURATION
V. LOCATION

OBJECTIVES
- Recognize the different types of muscles
- Understand the configuration of muscles

I. INTRODUCTION :

Myology is the branch of anatomy that studies the muscles. A muscle is a soft tissue mainly composed of contractile cells, allowing body movements.

The human body comprises three types of muscular tissue:
- Skeletal muscle – responsible for voluntary movements
- Smooth muscle – responsible for involuntary movements
- Cardiac muscle

Muscles are primarily responsible for posture, locomotion, and the movement of internal organs, such as heart contraction and the peristaltic movement of food through the digestive system.

The naming of muscles depends on several factors:
- Location (e.g., subclavian muscle, located under the clavicle)
- Shape (e.g., serratus anterior)
- Structure (e.g., biceps, triceps)
- Fiber direction (e.g., oblique)
- Action (e.g., short supinator)
- Size or volume (e.g., small and large pectoral muscles)

II. CHARACTERISTICS

1. Properties

- Excitability and contractility: the ability to contract in response to a nerve or electrical stimulus.
- Elasticity: the ability to stretch.
- Tonicity: the ability to maintain a certain state of partial contraction or muscle tone.

2. Number

There are approximately 600 skeletal muscles in humans (about 620 in total). This number may vary due to the presence of accessory muscles. They are distributed as follows:
- 100 muscles for the thoracic limbs
- 100 for the pelvic limbs
- 170 for the head and neck
- 200 for the trunk
- 50 for organs and systems

3. Weight

Muscles represent about 40% of total body weight, depending on physical activity.

III. CLASSIFICATION

There are three types of muscular tissue:
- Red muscles: Striated – voluntary (somatic functions: movement and posture)
- White muscles: Smooth – involuntary (vegetative functions)
- Cardiac muscle: Striated – involuntary

A. Skeletal Muscles (Red, Voluntary)

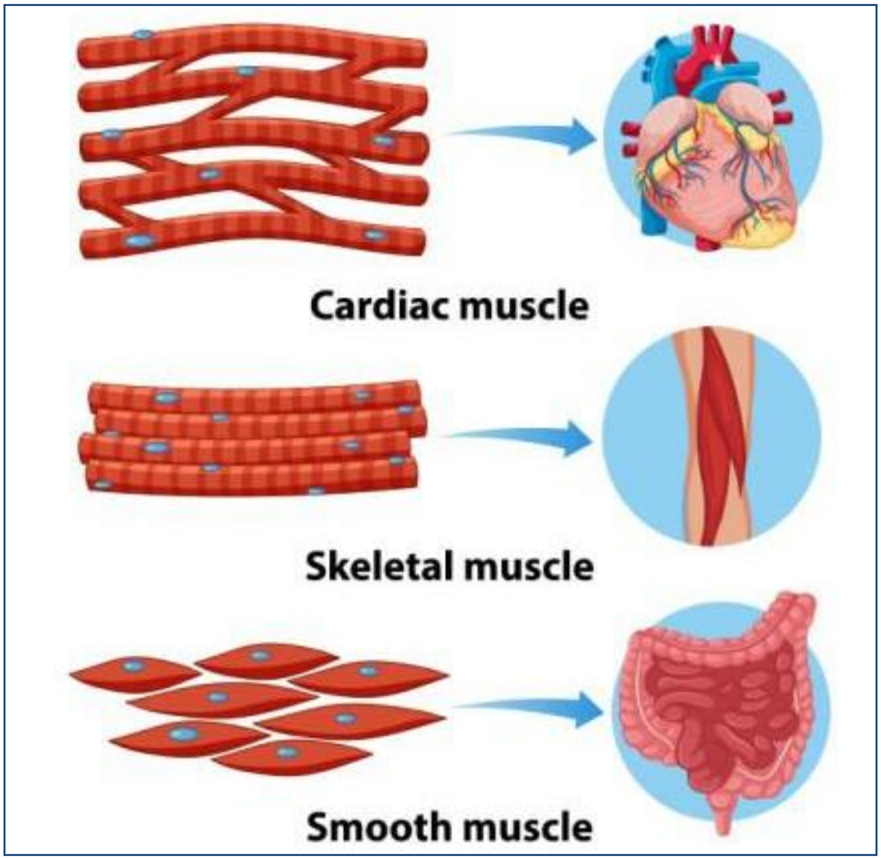
They are attached to bones via tendons and allow voluntary movements such as locomotion and posture maintenance.

B. **Smooth Muscles (Pale, Involuntary)**

Located in the walls of hollow organs such as the esophagus, stomach, intestines, bronchi, uterus, urethra, bladder, and blood vessels, forming a layer called the muscularis.
Their fibers contract slowly and independently of the will, under the control of the autonomic nervous system.

C. Cardiac Muscle (Myocardium)

Also known as the autonomous muscle, it resembles skeletal muscle in structure but is found only in the heart. It is fatigue-resistant, contracts regularly and automatically, and includes a mandatory relaxation period after each contraction. Its activity is independent of conscious control.



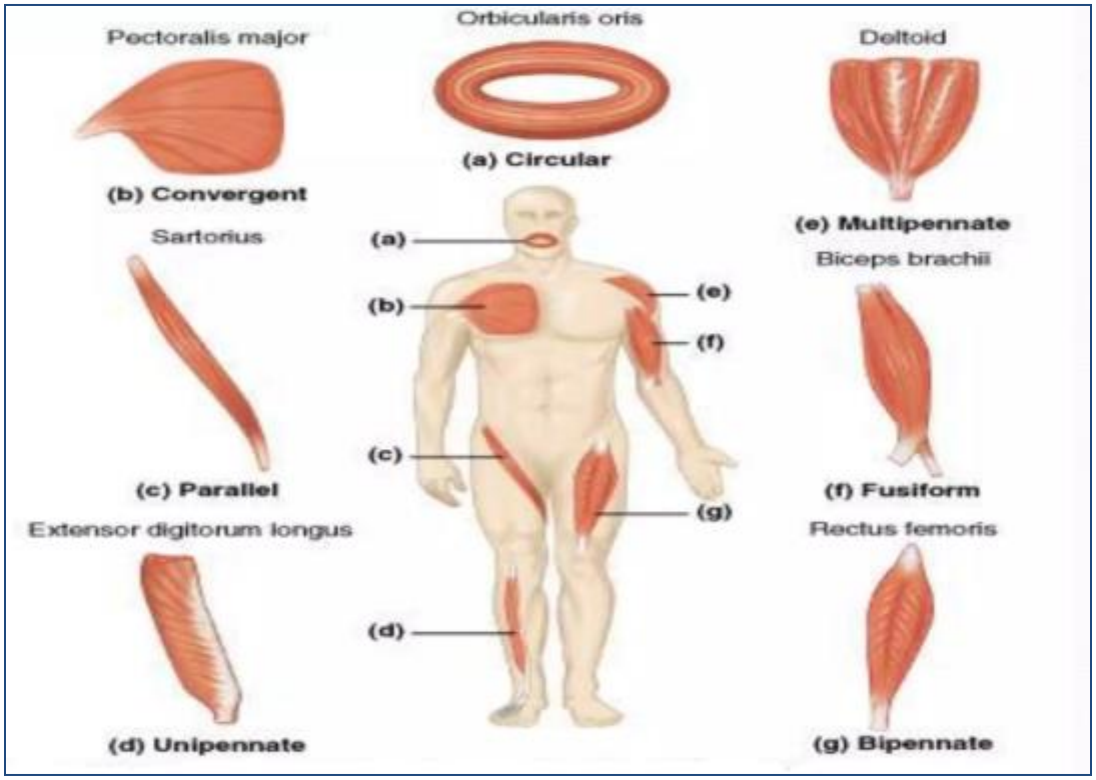
Type of muscles

IV. EXTERNAL CONFIGURATION

Muscles present to describe the muscular bodies and tendons and according to their shapes we distinguish several types of muscles

A. Muscle Body

- Long muscles: e.g., sartorius muscle
- Broad or flat muscles: fibers are parallel or fan-shaped, e.g., external oblique of the abdomen
- Short muscles: e.g., supraspinatus of the shoulder
- Annular or orbicular muscles: e.g., orbicularis oculi, sphincters
- Simple (monogastric) muscles: with a single belly, e.g., ulnar flexor of the carpus
- Compound (polygastric) muscles: with several bellies separated by an intermediate tendon, e.g., digastric of the neck, rectus abdominis
- Biceps, triceps, quadriceps: muscles with two, three, or four heads
- Multifid muscles: muscles whose distal ends terminate in several tendons, e.g., finger flexors and extensors.



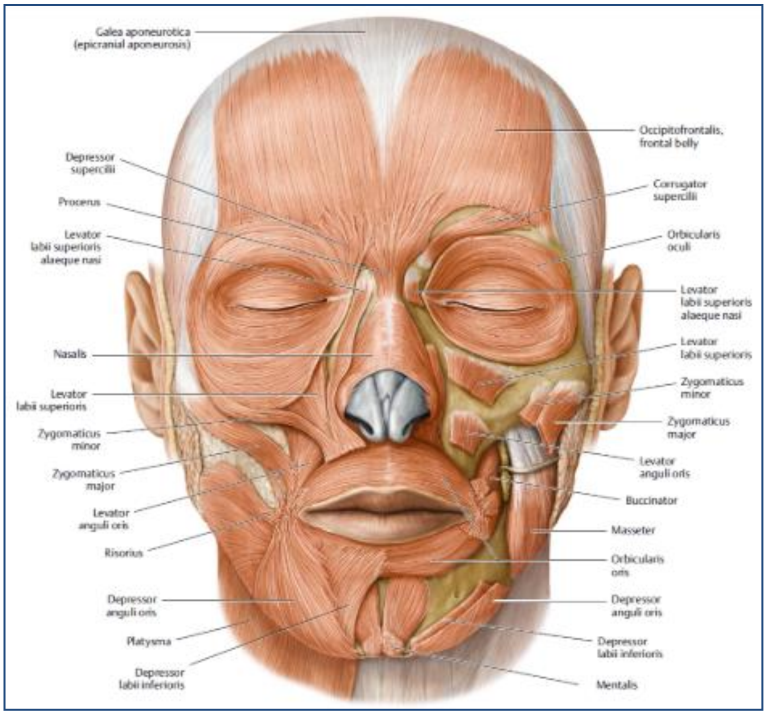
B. Tendons

Tendons are whitish fibrous structures that extend from muscle fibers and attach to bones, cartilages, or skin.

V. LOCATION

According to their position, muscles are classified as:

- Superficial or cutaneous muscles: located just beneath the skin (e.g., facial muscles)
- Deep or subaponeurotic muscles: located beneath the superficial fascia (e.g., muscles of the upper limb)



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